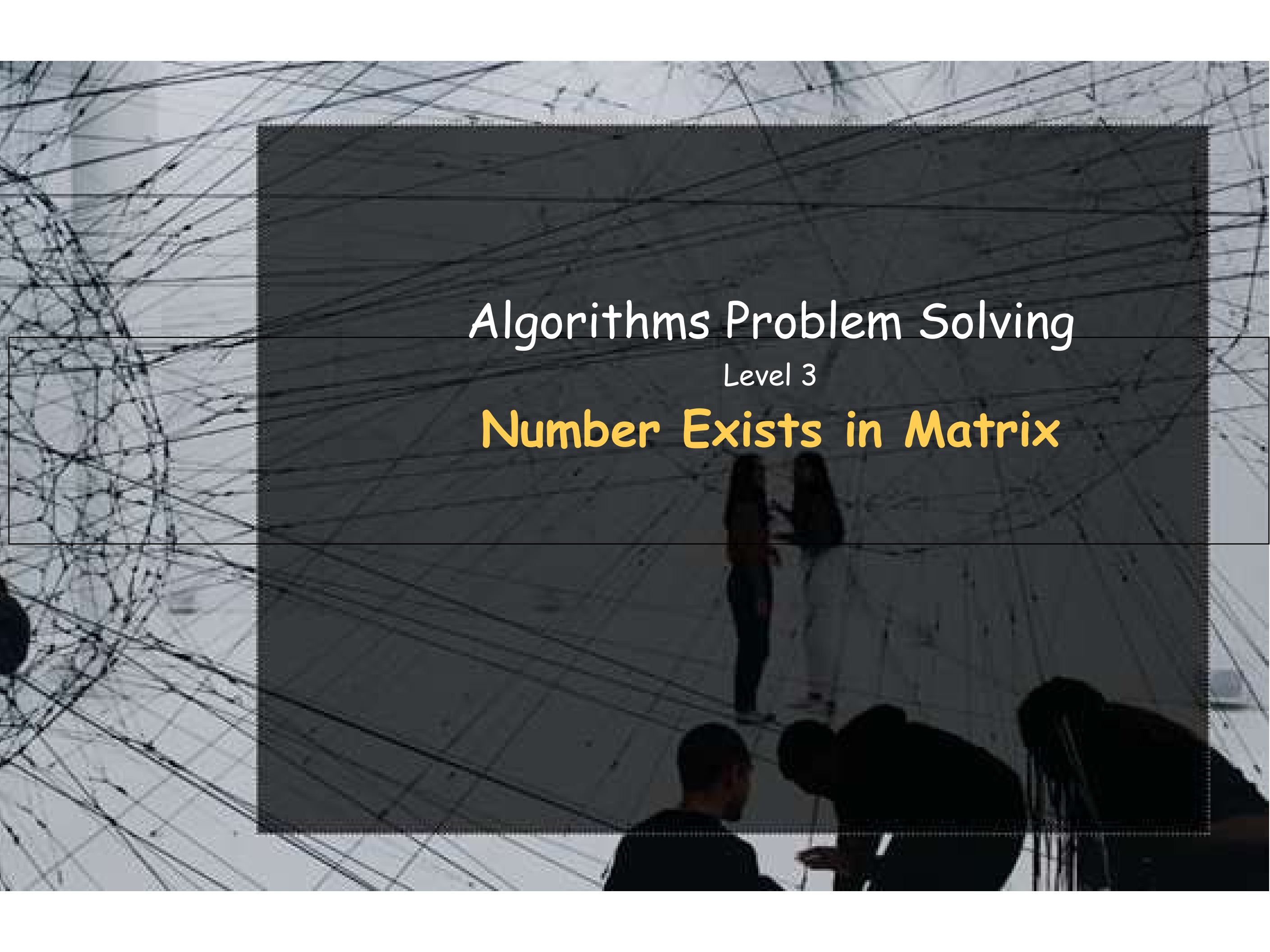


Algorithms Problem Solving

Level 3

Number Exists in Matrix



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Definition

Membership Search in 2D Matrix

 Program searches a matrix to determine whether a specific target number exists.

 The function returns a boolean result instead of counting all occurrences.

Explanation

Linear Search

-  The matrix is scanned row by row and column by column.

-  Each element is compared against the requested target.

Early Exit

-  As soon as a matching value is found, the function returns `true`.

-  Remaining elements do not need to be inspected.

Boolean Result

-  If the complete matrix is scanned without finding the value, the function returns `false`.

Procedures

1. Fill the matrix with values.
2. Ask the user for the target number.
3. Call `IsNumberInMatrix()`.
4. Traverse the matrix until a match is found.
5. Display the corresponding YES or NO message.

💡 Example

Matrix1:

10 20 30

15 25 35

27 29 37

Please enter the number to look for in the matrix ? 30

➡ Result

Yes it is there.

Note

- ◆ This is a linear search because the matrix is not assumed to be sorted.
- ◆ Best-case time complexity is $O(1)$ when the target is found at the first position.
- ◆ Worst-case time complexity is $O(N \times M)$ when the target is absent or appears at the last position.

⊘ Warning

- ◆ Do not continue scanning after finding a match when only existence is required.
- ◆ Passing a large matrix by value creates an unnecessary copy; prefer ``const`` reference where appropriate.
- ◆ An unsorted matrix generally requires a complete scan in the worst case.



Comparison

▣ `IsNumberInMatrix()`

- ✓ Answers whether a value exists.
- ✓ Can stop immediately after the first match.
- ✓ Returns a boolean.

▤ `CountNumberInMatrix()`

- ✓ Determines the exact number of occurrences.
- ✓ Must continue scanning after finding matches.
- ✓ Returns an integer count.

✦ Summary

- ✓ Performs a linear membership search across a matrix.
- ✓ Uses early exit when the target is found.
- ✓ Returns a boolean existence result.
- ✓ Avoids unnecessary scanning when the answer is already known.

Thank you 😊